

✓ SOP #3 – Troubleshooting Techniques for Chillers

Applies to: Centrifugal, Screw, Scroll, Absorption, and Modular Air- or Water-Cooled Chillers

Trigger Events: System trips, poor cooling performance, alarms, vibration, or poor EER/COP readings

🔧 1. Common Failure Signs & Diagnoses

Symptom	Possible Cause	Recommended Action
High condenser pressure	Poor water flow, fouled tubes, air in system	Backflush tubes, check pump & tower setpoints
Low suction pressure	Low refrigerant charge, TXV issue	Leak test & inspect metering device
Frequent compressor starts/stops	Oversized system, control error	Adjust setpoints, inspect economizer
Discharge line too hot	Non-condensables, oil imbalance	Pull vacuum & analyze oil quality
Evaporator freeze risk	Poor flow, sensor malfunction	Verify water flow switch & sensor inputs
Fault code "Low Delta-T"	Chilled water bypassing, fouled heat exchanger	Check valves, flow, and clean tubes